

1st SIT EXAMINATION QUESTION PAPER:**Year Long 2022**

Module Code:	CC6001NI
Module Title:	Advanced Database System Development
Module Leader:	Mr. Rohit Panday (Islington College)

Date:	
Day / Evening:	Day
Start Time:	
Duration:	2 Hours

Exam Type:	Unseen Examination
Materials supplied:	NA
Materials permitted:	NA
Warning:	Candidates are warned that plagiarism is a serious assessment offence.

Instructions to candidates:	Please Note: This examination paper consists of 6 pages, inclusive of this cover page. The student must complete four (4) out of five (5) questions. This examination accounts for 40% of your total module grades. You are to return this examination paper BEFORE you leave the examination room. DO NOT TURN THE PAGE OVER UNTIL INSTRUCTED
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Attempt Any 4 Questions

Question no 1.

a) Highlight the major difference between DDL, DML and DCL with one example for each.

[6 marks]

b) Define all three types of anomalies. Normalize the following table up to 3NF showing all the normal forms and dependencies. (Mention all assumptions that have been made during normalization)

[6 marks + 13 marks]

E_id	E_name	E_dob	E_hire_date	D_id	D_name	Salary	Loc_id	Loc_name	job_status
E1	Raman Singh	1997-May-17	2018-sep-11	Dep1	Academics	30000	Loc1	Kamalpokhari	part-time
E2	Rajiv Shakya	1998-Jan-2	2019-jan-12	Dep1	Academics	40000	Loc1	Kamalpokhari	full-time
E3	Jeeban Raj Bhat	1997-feb-20	2016-feb-11	Dep1	Academics	35000	Loc1	Kamalpokhari	part-time
E3	Jeeban Raj Bhat	1997-feb-20	2016-march-11	Dep2	IT	30000	Loc2	Kalanki	full-time

Fig. employee's allocated department details

Question no 2.

The following relational schema specifies a part of a relational database for an organization's information system:

Employee (E#, E_Name, Gender, DOB, Address, Tel, D#)

Department (D#, D_Name, L#)

Location (L#, Street, Zone, Country)

Job (E#, D#, Job_title, Salary)

The following assumptions are made:

- Employee job for any department is specified by the employee number (S#), department number (M#), job_title (e.g., 'manager', or 'Assistant'), and salary.

a) Express each of the following queries firstly in SQL and secondly in Relational Algebra:

(i) Find the names of all the department where people from Kathmandu is working.

[5 Marks]

(ii) Find the list of country names where "Mr. David" is working as 'Manager'.

[6 Marks]

b) calculate the total number of tuples for the query execution while using the operation (**SELECT – JOIN – PROJECT**)

The following assumptions are made:

- 100 records in employee table, 2 employees with name 'Mr. David', 50 employees with address as 'Kathmandu',
- 10 Department record in department table and 50 record in location table
- 10 records in job table where 1 record as 'Manager' for job title.
- Only results (intermediate relations) of up to 100 tuples can be kept in memory during query processing

i) Find the names of all the department where people from Kathmandu is working.

[6 Marks]

(ii) Find the list of country names where "Mr. David" is working as 'Manager'.

[8 Marks]

Question no 3.

a) Define the locking mechanism. Explain locking types and their protocol.

[2 Marks + 3 Marks + 6 Marks]

b) The following list represents the sequence of events in an interleaved execution of a set of transactions T1, T2, ... T24 in a concurrency system based on locking, where A, B, ...H are data items. Assume that FETCH A acquires an S lock on A, UPDATE A promotes that lock to an X lock, and all locks are held to the next synch point.

<i>Time</i>	<i>Transaction</i>	<i>operation</i>
<i>t0</i>	T1	<i>FETCH A</i>
<i>t1</i>	T2	<i>FETCH B</i>
<i>t2</i>	T2	<i>FETCH C</i>
<i>t3</i>	T3	<i>FETCH B</i>
<i>t4</i>	T5	<i>FETCH A</i>
<i>t4</i>	T5	<i>UPDATE A</i>
<i>t5</i>	T5	<i>FETCH D</i>
<i>t6</i>	T7	<i>FETCH E</i>
<i>t7</i>	T8	<i>FETCH G</i>
<i>t8</i>	T1	<i>COMMIT</i>
<i>t9</i>	T7	<i>FETCH A</i>
<i>t10</i>	T2	<i>UPDATE C</i>
<i>t11</i>	T3	<i>UPDATE B</i>
<i>t12</i>	T8	<i>FETCH E</i>
<i>t13</i>	T8	<i>UPDATE E</i>
<i>t14</i>	T8	<i>UPDATE G</i>
<i>t15</i>	T7	<i>ROLLBACK</i>
<i>t16</i>	T2	<i>FETCH G</i>
<i>t17</i>	T10	<i>FETCH C</i>
<i>t18</i>	T8	<i>FETCH A</i>
<i>t19</i>	T6	<i>FETCH F</i>
<i>t20</i>	T9	<i>FETCH F</i>
<i>t21</i>	T9	<i>UPDATE F</i>
<i>t22</i>	T3	<i>FETCH K</i>
<i>t23</i>	T3	<i>UPDATE K</i>
<i>t24</i>	T5	<i>FETCH K</i>
<i>t25</i>		

(i) Provide a Wait-for-Graph to illustrate whether there is any deadlock at time *t25*. You are advised to provide full and appropriate workings.

[10 Marks]

(ii) Discuss how the system could recover if it were deadlocked at time *t25*, and justify your choice of victim transaction(s).

[4 Marks]

Question no 4.

a) Define Transaction and its properties. Discuss the purpose of database recovery and outline the techniques to Recover the Database from any disaster.

[5 marks + 5 Marks]

b) Discuss the concept of:

- i) Candidate Key
- ii) Distributed Database
- iii) Centralized Database
- iv) Data Replication

[6 Marks]

(c) In the course of transaction processing, the system takes checkpoints at specified intervals. Assume that sometimes after a particular checkpoint was taken a system failure occurred. Consider the following transactions in the scenario:

T1 started before the checkpoint but did not complete when the system failed.
T2 started before the checkpoint and was completed before the system failed.
T3 started after the checkpoint and was completed before the system failed.
T4 started and was completed before the checkpoint was taken.
T5 started after the checkpoint but did not complete when the system failed.

(i) Draw a diagram to illustrate the above set of transactions.

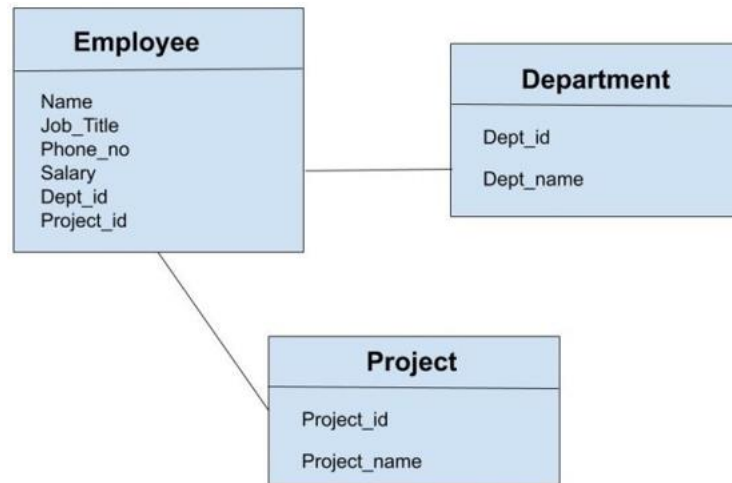
[5 Marks]

(ii) For each of the above transactions, indicate which must be redone, which must be undone and which is unaffected by the failure.

[4 Marks]

Question no 5

a) Considering the following oracle schema with username **DBOWNER**.



i) Write down a SQL query to create 2 users with username, DBReader and DBWriter.

[4 marks]

ii) Grant select permission for both users on any table from the above schema.

[4 marks]

iii) Grant insert, delete and update permission for DBWriter user on all table from the above schema and provide insert permission for DBReader on employee and department table.

[7 marks]

b) Create **Access matrix** for **CRUD** operation in tabular form from the above provided permission for both DBReader and DBWriter Role.

[10 Marks]

-- End of Question Paper --